International Trade Theory and Development Strategy

Chapter 12

Econ 322 – The Global Economy: Trade and Development

University of Tennessee, Knoxville

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Based on Todaro, P. M. and Smith, C. S. (2020), *Economic Development*, 13th ed., Pearson.

12.1 Economic Globalization: An Introduction

- **Globalization**—many interpretations
- Core economic topics—the **increased openness of economies to international trade, financial flows, and foreign direct investment**
- Concerns with globalization have centered around the **unevenness of the process and risks**
- The World Trade Organization (**WTO**): Geneva-based watchdog and enforcer of international trade agreements since 1995

Centre William Rappard, Geneva—headquarters of the WTO (and of the GATT before it).

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Cases of Development and Trade

- **China, India, and South Korea** are all cases of reduced international trade barriers leading to fast growth patterns.
- These countries specialized in exporting industries that allowed them to benefit from globalization.
- However, trade came with costs: inequality has been accentuated among these countries, for example in between coastal and inland China.

Three routes to export-led growth. *China*: Special Economic Zones and manufacturing exports after 1978 ("factory of the world"). *India*: liberalization in 1991, then IT services and outsourcing. *South Korea*: state coordination with large conglomerates (chaebols), moving from textiles to ships, cars, and semiconductors. In each case gains were uneven across regions and groups.

Trade Agreements

- The trade rules negotiated under the auspices of the WTO (1995) are key examples of how rules of the game are being created.
- Trade protectionism as practiced by developed countries tends to fall most heavily on the poorest developing countries because developed-country protection focuses on agriculture.
- Tariffs placed by developed countries on imports from developing countries were, by 2010, about twice those placed on imports coming from other developed countries.

Trade Agreements

- Trade agreements may attempt to "level the playing field" for poor countries. What does this mean?
- International and national actions to **prevent**:
  - Fueling corrupt governments that violate human rights.
  - Financing violent/exploitative rebel groups that profit from trade. (Diamonds and oil)
  - Trade of illegal goods and narcotics.
  - Codes of conduct for multinational corporations (political and other types of behavior that hamper developing countries).
  - Reasoning limits to property rights (provision of life-saving medicine in LICs that can't afford monopolistic rents).

Some examples of negative trade practices

- Oil Producing and Coffee Producing areas in Colombia are geographically different.
- The country also suffered intense guerrilla warfare and terrorism from 1964 to 2016 (and, less intense, today, once again).
- Dube and Vargas (2013) show that when the price of oil goes up, terrorist attacks increased.
- When the price of coffee goes up, terrorist attacks decrease.
- Organized groups steal production from pipelines and sell it illegally abroad, financing their rebellion.
- Increases in coffee prices raise the opportunity cost of guerrillas, discourage war.

Source: Dube and Vargas (2013)

Banana Republics

- The United Fruit Company, an American multinational corporation, was infamous across Costa Rica, Honduras, and Guatemala.
- It held the monopoly in the international trade of bananas. They discouraged the government from building highways, to eliminate competition to their railroads.
- It was well known for exploitation of workers, bribes, tax avoidance, and lobbying US military interventions across LATAM.
- Between 50 and 3,000 UFCO union workers were massacred in Santa Marta, Colombia by the military.
- In Honduras, early in the 1900s, investors in the banana trade sailed from New Orleans in an attempt coup de etat. Before violence ensued, the president was forced to step down. The new President rapidly granted exclusive railroad concessions to fruit companies for up to 99 years.

But “levelling the field” can mean other things

- Businesses will quickly attempt to use international trade regulation to their advantage.
- Protectionism stresses the “levelling the playing field” argument, quickly transforming into pervasive tariffs and ISI.
- Protectionism is hard to remove. Once firms are protected, productivity is not granted.
- Thus, more protections and subsidies are required to keep them afloat.

12.2 International Trade: Some Key Issues

- Developing countries rely heavily on exports of **primary products** with attendant economic and political economy risks and uncertainty, and evidence of **deteriorating terms of trade**
- Developing countries rely heavily on **imports of machinery, capital goods, and intermediate producer goods,** and in some cases, consumer necessities
- Many developing countries have **chronic deficits on current and capital accounts**, which deplete their reserves, cause currency instability, and may slow economic growth
- Since the 1990s, a larger number of developing countries have sought to promote exports and accumulate large **foreign exchange reserves to cushion against crises**, spurring new policy debates

12.2 International Trade: Some Key Issues (Continued)

- Five Basic Questions about Trade and Development
  - How does international trade affect **economic growth**?
  - How does trade alter the **distribution of income**?
  - How can trade **promote development**?
  - Can developing countries determine **how much they trade**?
  - Is an **outward-looking or an inward-looking** trade policy best? (Or can hybrids of these be effective?)

12.2 International Trade: Some Key Issues (Continued)

- Importance of exports to different developing nations
- Exports of developing countries are generally **less diversified** than those of developed countries
- **Merchandise exports** as a share of GDP are often **higher** for developing countries

Why this matters. Concentrated exports are like holding a single stock: when the price of that one good falls, national income falls with it. And because domestic markets are small, developing countries depend more on world demand (Vietnam's merchandise exports are near 100% of GDP; the U.S. figure is under 10%, Table 12.1), so global slowdowns hit them harder.

Table 12.1 Structure of Merchandise Exports: Selected Countries, 2017 (1/2)

Country	Code	GDP, $ bn	Merch. exports, $ bn	Merch. exports, % GDP	Food, % of total	Agric. raw mat., % of total	Fuels, % of total	Ores & metals, % of total	Manuf., % of total
Algeria	DZA	167.6	35.2	21%	1	0	95	0	4
Benin	BEN	9.2	2.0	21%	31	50	3	1	16
Bolivia	BOL	37.5	7.8	21%	17	0	38	40	5
Brazil	BRA	2053.6	217.8	11%	36	4	9	12	38
Burkina Faso	BFA	12.3	2.9	23%	37	36	1	17	9
Burundi	BDI	3.2	0.2	5%	80	0	0	8	12
Central African Republic	CAF	2.2	0.1	6%	1	12	0	4	65
China	CHN	12143.5	2263.3	19%	3	0	2	1	94
Cote d'Ivoire	CIV	38.1	11.9	31%	60	10	13	1	16
Ecuador	ECU	104.3	19.1	18%	50	6	37	1	6
Egypt, Arab Rep.	EGY	235.4	25.6	11%	19	2	21	4	54
Gambia, The	GMB	1.5	0.1	7%	77	1	0	0	22
Ghana	GHA	59.0	13.8	23%	40	3	43	3	11
India	IND	2652.6	299.3	11%	12	1	12	4	71
Indonesia	IDN	1015.4	168.8	17%	23	5	22	6	44

Source: World Bank World Development Indicators, Table 4.4, accessed 6 August 2019.

Table 12.1 Structure of Merchandise Exports: Selected Countries, 2017 (2/2)

Country	Code	GDP, $ bn	Merch. exports, $ bn	Merch. exports, % GDP	Food, % of total	Agric. raw mat., % of total	Fuels, % of total	Ores & metals, % of total	Manuf., % of total
Iran, Islamic Rep.	IRN	454.0	92.8	20%	6	0	71	3	20
Japan	JPN	4860.0	698.1	14%	1	1	2	2	88
Malawi	MWI	6.3	0.9	14%	90	2	0	0	8
Malaysia	MYS	314.7	217.7	69%	11	2	15	4	68
Mauritius	MUS	13.3	2.3	18%	38	1	2	0	57
Mexico	MEX	1158.1	409.4	35%	8	0	6	3	82
Mozambique	MOZ	12.7	4.7	37%	10	1	52	31	6
Nicaragua	NIC	13.8	5.2	37%	52	1	0	1	46
Nigeria	NGA	375.7	44.5	12%	2	0	96	0	2
Peru	PER	210.7	45.3	21%	23	1	9	55	11
Philippines	PHL	313.6	68.7	22%	9	1	2	6	83
Russian Federation	RUS	1578.6	353.5	22%	6	3	59	6	22
South Africa	ZAF	348.9	88.8	25%	11	2	13	26	47
United Kingdom	GBR	2637.9	441.1	17%	7	1	8	4	77
United States	US	19485.4	1546.3	8%	10	2	11	3	62
Vietnam	VNM	223.8	214.3	96%	12	1	2	1	83

Source: World Bank World Development Indicators, Table 4.4, accessed 6 August 2019.

Demand Elasticities and Export Earning Instability

- Often, low-price elasticity of demand for agricultural commodities, but supply shocks
- Often low-price elasticity of supply for basic commodities, but demand shocks
- Result can be export earnings instability, risks to income
- Also, low-income elasticity of demand for primary products

The Terms of Trade and the Prebisch-Singer Hypothesis

- Total export earnings depend upon:
  - **Total volume** of exports sold; and
  - **Price** paid for exports
- Prebisch and Singer argued that **commodity export prices fall over time**, so developing countries lose revenue unless they can continually increase export volumes
- They concluded that developing countries need to **avoid dependence on primary exports**
- Some evidence suggests that relative prices within manufactures are also diverging with falling prices for low-skill products

Raúl Prebisch (1901–1986), Argentine economist, head of ECLA and first Secretary-General of UNCTAD.

Wikimedia Commons, Public domain

12.3 The Traditional Theory of International Trade

- Comparative advantage and specialization
- Relative factor endowments and international specialization: the Neoclassical model
  - Ricardo and Mill (static model)
  - Heckscher-Ohlin (factor endowment theory): different products require productive factors in different ratios, and countries have different endowments of factors of production

David Ricardo (1772–1823), who formalized comparative advantage in 1817.

Wikimedia Commons, Thomas Phillips, Public domain

Comparative advantage

- **A scenario** –Two sailors are stranded on an island. To survive, they need to undertake basic economic activities, such as fishing, cooking, and building shelters. One sailor is much younger and better at all the activities than the other.
- **Questions:**– Should they try to survive independently on their own, or divide the activities between them and work in collaboration?– If working together is in the better interest of both sailors, then how should they divide their work?

Comparative advantage

- **Labor Productivity** – The number of units of output produced in an hour of labor, or the number of hours of labor required to produce one unit of output (*unit labor requirement*, ULR).

Production hours	Clothing	Computers
USA	0.8	2.0
China	1.0	4.0

Comparative advantage

Absolute advantage: it takes fewer hours for a person (a country) to perform a task than another.
Comparative advantage: a person's (a country's) opportunity cost of performing a task is lower than that of another.

- Who has an absolute advantage in updating websites?
- Who has an absolute advantage in repairing bikes?
- What are the opportunity costs of undertaking each activity for Paula and for Beth?

Production time	Web update	Bike repair
Paula	20 minutes	10 minutes
Beth	30 minutes	30 minutes

Comparative advantage

Production time	Web update	Bike repair
Paula	20 minutes	10 minutes
Beth	30 minutes	30 minutes

Opportunity cost	Web update	Bike repair
Paula	2 repairs	0.5 update
Beth	1 repair	1 update

Red: the activity in which each person has the **comparative advantage** (lower opportunity cost).

Comparative advantage

Hourly output	Web update	Bike repair
Paula	3 updates	6 repairs
Beth	2 updates	2 repairs

In an 8-hour workday with an order of 16 web updates:

- **Scenario A:** Paula spends half of her time on each activity: 12 updates and 24 repairs; then, Beth produces 4 updates and 12 repairs. So, the total output is 16 updates and 36 repairs.
- **Scenario B:** Specialization: Paula produces 48 repairs and Beth produces 16 updates.

Comparative advantage

- Specialization gives **12 more repairs for the same inputs!**

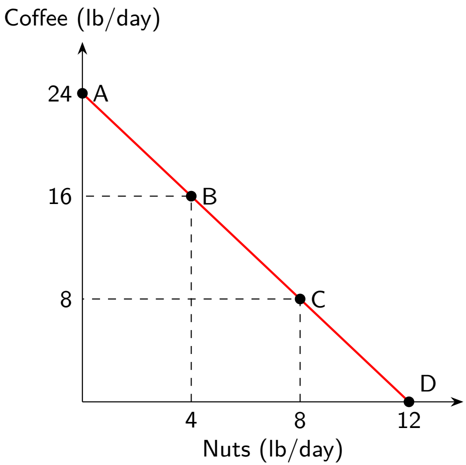
The principle of Comparative Advantage – Everyone is better off when each concentrates on the activity with the lowest opportunity cost.

- Specialize in producing the goods in which one has a comparative advantage;
- Exchange goods with one another;
- Everyone would enjoy more goods than they would otherwise if they produce everything on their own.

Comparative advantage

Susan's productivities (ULR)	
Coffee	Nuts
0.25 hrs/pound	0.5 hrs/pound

- 6 working hours per day;
- **PPC** is a graph that describes the possible output combinations of two goods that can be produced with given resources.



Comparative advantage

- **Susan's Opportunity Cost**
  - Opp. cost of producing one pound of nuts equals

$$\frac{0.5 \text{ hrs}}{0.25 \text{ hrs}} = 2 \text{ pounds of coffee}, \quad \frac{\text{Loss in coffee}}{\text{Gain in nuts}} = \text{slope of PPC} = \frac{24}{12} = 2$$

  - Opp. cost of producing one pound of coffee equals

$$\frac{0.25 \text{ hrs}}{0.5 \text{ hrs}} = \frac{1}{2} \text{ pound of nuts}, \quad \frac{\text{Loss in nuts}}{\text{Gain in coffee}} = \text{reciprocal of PPC slope} = \frac{12}{24} = \frac{1}{2}$$

Comparative advantage

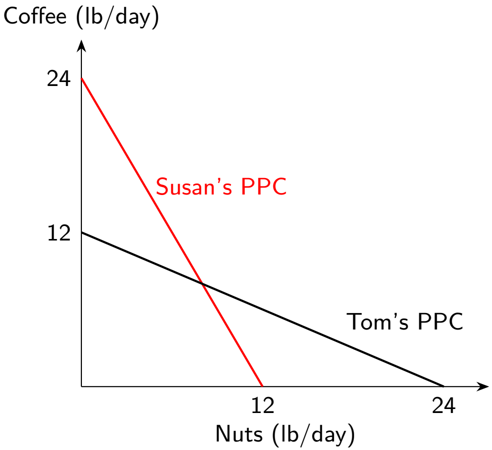
- Introducing Tom who also works 6 hours a day

Production time	Coffee (pound)	Nuts (pound)
Susan	0.25 hrs	0.5 hrs
Tom	0.5 hrs	0.25 hrs

Opportunity cost	Coffee (pound)	Nuts (pound)
Susan	0.5 pds of nuts	2 pds of coffee
Tom	2 pds of nuts	0.5 pds of coffee

Comparative advantage

- Susan's comparative advantage is on **coffee** production.
- Tom's comparative advantage is on **nuts** production.

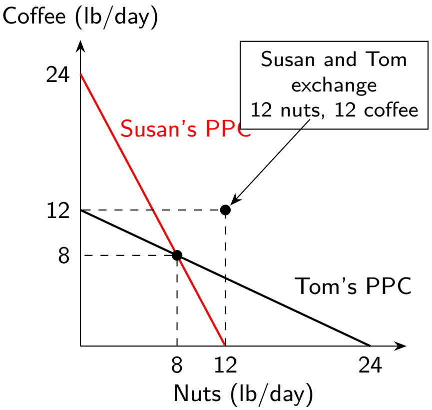


Comparative advantage

- **Gains from specialization and trade**
  - Without specialization and trade, Susan and Tom have to produce both coffee and nuts on their own;
  - On her own, Susan would be willing to give up 2 pounds of coffee for 1 pound of nuts;
  - On his own, Tom would be willing to give up 2 pounds of nuts for 1 pound of coffee.

Room for a deal. Susan values 1 pound of nuts at up to 2 pounds of coffee; Tom can make a pound of nuts for only 0.5 pound of coffee. Any exchange rate between 0.5 and 2 pounds of coffee per pound of nuts leaves both better off—the next slide uses 1:1.

Comparative advantage



- With specialization and trade, would both Susan and Tom be better off if
  - Susan specializes in producing coffee;
  - Tom specializes in producing nuts;
  - And, Susan trades **one** pound of coffee for **one** pound of nuts with Tom.
- Without specialization and trade, the total output and consumption is 16 pounds of coffee and 16 pounds of nuts;
- With specialization and trade, the total output and consumption is 24 pounds of coffee and 24 pounds of nuts.

12.4 Critique of Traditional Free-Trade Theory (1/2)

- The following assumptions of the basic **Neoclassical model** have been scrutinized:
  - **Fixed resources, full employment, international factor immobility**
  - **Fixed, freely available technology** and **consumer sovereignty vs. product cycle**, ongoing development of synthetic substitutes for developing countries' exports, opportunities for dynamic gains in leading sectors
  - **Internal factor mobility** vs. **different types of structural rigidities**; and **perfect competition** vs. **pervasive market power**
  - **Governmental non-interference in trade** vs. **active trade policies**
  - **Balanced trade** and **international price adjustments** vs. **instability**
  - Trade gains accruing to nationals vs. export enclaves with foreign ownership; distinction between **GDP and GNI** becomes important

12.4 Critique of Traditional Free-Trade Theory (2/2)

- Fixed Resources, Full Employment, and the International Immobility of Capital and Skilled Labor
  - Challenged by **North-South trade models**
- Porter's "Competitive Advantage" theory:
  - Traditional trade theory applies only to **basic factors** (unskilled labor, physical resources)
  - But the creation of **advanced factors** (knowledge resources, specialized infrastructure) is the priority
  - Central task to "escape from the straitjacket of factor-driven national advantage"

Alternative Theories: Vent for Surplus (Figure 12.2)

- Vent-for-surplus theory (Adam Smith; developed by Hla Myint for developing countries)
- Traditional theory assumes full employment; but a closed economy may have idle land and labor
- Opening to trade provides an outlet (a "vent") for these unused resources: production moves from V to B and exports pay for imports
- Gains come from *activating* idle resources, not from reallocating fully employed ones

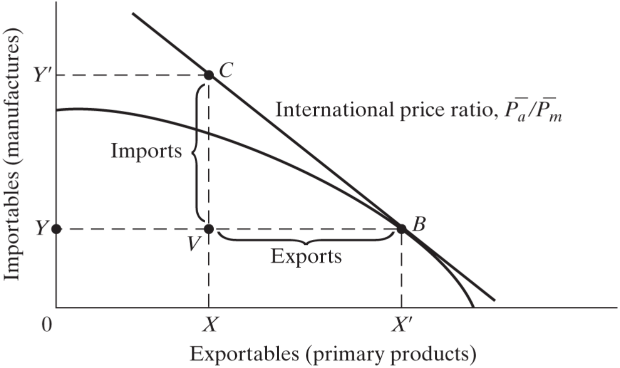


Figure 12.2 The vent-for-surplus theory of trade.

Source: Todaro and Smith (2020), Economic Development, 13th ed.

12.5 Export Promotion versus Import Substitution (1/3)

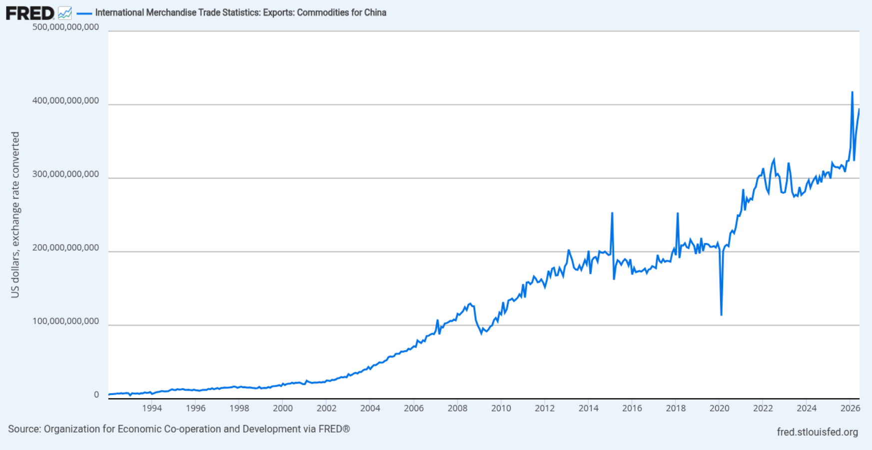
- Export promotion: looking outward and seeing trade barriers
- Primary-commodity export expansion faces **limited demand**:
  - Low-income elasticities
  - Low population growth rates in developed economies
  - Decline in prices implies low revenue (some periods of price spikes, including recent years, but very long-run trend has been downward)
  - Lack of success with international commodity agreements
  - Development of synthetic substitutes
  - Agricultural subsidies
- Primary-commodity export expansion also faces **supply rigidities**

12.5 Export Promotion versus Import Substitution (2/3)

- Expanding exports of manufactured goods: greater successes, particularly China; unevenly distributed across the developing world
- Import substitution: looking inward but still paying outward
  - Tariffs, infant industries, and the theory of protection

"Looking inward but still paying outward." Behind tariff walls, countries still had to import machinery, parts, and technology for the protected industries—often increasing debt and dependence even while pursuing self-sufficiency. The infant-industry argument works only if protection is temporary, like training wheels.

China's export boom



China's monthly merchandise exports in U.S. dollars, 1992–2026: the clearest case of successful manufactured-export expansion.

Source: FRED, Federal Reserve Bank of St. Louis (series XTEXVA01CNM667S)

12.5 Export Promotion versus Import Substitution (3/3)

- The import substitution (IS) industrialization strategy and results
  - Protected industries get inefficient and costly
  - Foreign firms often benefit more
  - Subsidization of imports of capital goods tilts pattern of industrialization and contributes to balance of payments (BOP) problems
  - Overvalued exchange rates hurt exports
  - Does not stimulate self-reliant integrated industrialization

12.6 South-South Trade and Economic Integration

- Economic Integration: Theory and Practice
  - The growth of trade among developing countries.
  - Integration encourages rational division of labor among a group of countries and increases market size
  - Provides opportunities for a coordinated industrial strategy to exploit economies of scale
  - Trade creation
  - Debate on trade diversion

12.6 South-South Trade and Economic Integration (Continued)

- Regional trading blocs and the globalization of trade
- Examples:
  - Canada-Mexico-U.S. (formerly known as NAFTA)
  - European Union
  - MERCOSUR
  - SADC
  - ASEAN
  - Local conditions matter
- Not fully answered: What effects do blocs promote have in promoting or slowing (or modifying) globalization?
- Can blocs, including among developing countries, mitigate potential negative impacts of globalization?

12.7 Trade Policies of Developed Countries

- Rich-nation economic and commercial policies matter for developing countries
  - Tariff and *non-tariff* barriers to developing country exports
  - Adjustment assistance for displaced workers
  - General impact of economic policy
- World Trade Organization
- Despite eight liberalization rounds over 50 years, trade barriers remain in place in agriculture; and, through various mechanisms, to a degree in other sectors
- Doha Development Round begun 2001 were to change the nominal focus to needs of developing world; but talks remained stalled over the next two decades, passing self-imposed deadlines

Chapter summary

- Globalization means greater openness to trade, financial flows, and FDI; China, India, and South Korea grew fast through exports, but gains were uneven
- Developing countries depend on primary exports, face volatile earnings and (per Prebisch–Singer) declining terms of trade, and often run chronic current-account deficits
- Comparative advantage: specialize where opportunity cost is lowest and trade—everyone can consume more (Susan and Tom: 16 to 24 pounds of each good)
- The neoclassical model's assumptions (full employment, fixed technology, immobile factors, no government) are questioned for developing countries; vent-for-surplus offers an alternative
- Export promotion has outperformed import substitution, which bred inefficient protected industries; South–South integration and reform of rich-country protection (WTO, Doha) remain on the agenda

Concepts for Review

- Absolute advantage
- Autarky
- Balanced trade
- Barter transactions
- Capital account
- Commodity terms of trade
- Common Market
- Comparative advantage
- Current account
- Customs Union
- Depreciation
- Devaluation
- Dual exchange rate

Concepts for Review (Continued)

- Economic Integration
- Economic Unions
- Effective Rate of Protection
- Enclave economies
- Exchange Control
- Export dependence
- Export earnings instability
- Factor endowment trade theory
- Factor price equalization
- Flexible exchange rate
- Foreign-exchange earnings
- Free market exchange rate
- Free trade

Concepts for Review (Continued)

- Free trade area
- Gains from trade
- General Agreement on Tariffs and Trade (GATT)
- Globalization
- Growth poles
- Import substitution
- Income elasticity of demand
- Increasing returns
- Industrialization Strategy Approach
- Industrial policy
- Infant industry
- International commodity agreements
- Inward-looking development policies

Concepts for Review (Continued)

- Managed float
- Monopolistic market control
- Multifiber Arrangement (MFA)
- New protectionism
- Nominal rate of protection
- Nontariff trade barriers
- North-South trade models
- Official exchange rate
- Oligopolistic market control
- Outward-looking development policies
- Overvalued exchange rate
- Parallel exchange rate
- Prebisch-Singer thesis

Concepts for Review (Continued)

- Price elasticity of demand
- Primary products
- Product Cycle
- Product differentiation
- Quotas
- Regional trading bloc
- Rent
- Returns to scale
- Risk
- Specialization
- Subsidy
- Synthetic substitutes
- Tariffs

Concepts for Review (Continued)

- Trade creation
- Trade deficits
- Trade diversion
- Trade liberalization
- Trade optimists
- Trade pessimists
- Uncertainty
- Undervalued exchange rate
- Uruguay Round
- Value added
- Vent-for-surplus theory of international trade
- Wage-price spiral
- World Trade Organization (WTO)